

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/Mall_Customers.csv')
```

```
df.head()
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Next steps:

[Generate code with df](#)

[View recommended plots](#)

[New interactive sheet](#)

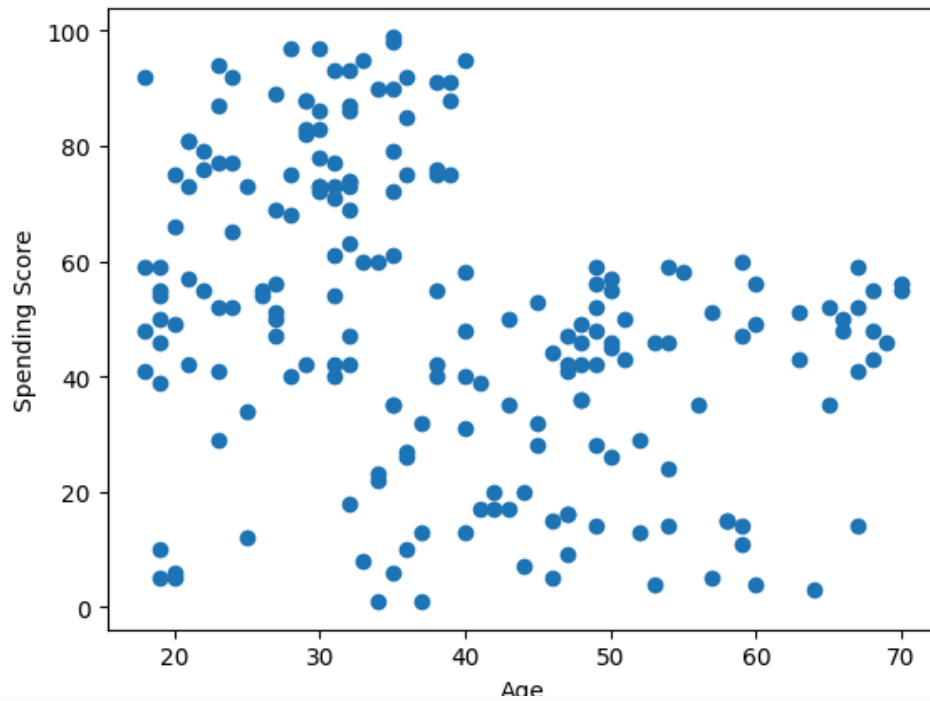
```
df.shape
```

 (200, 5)

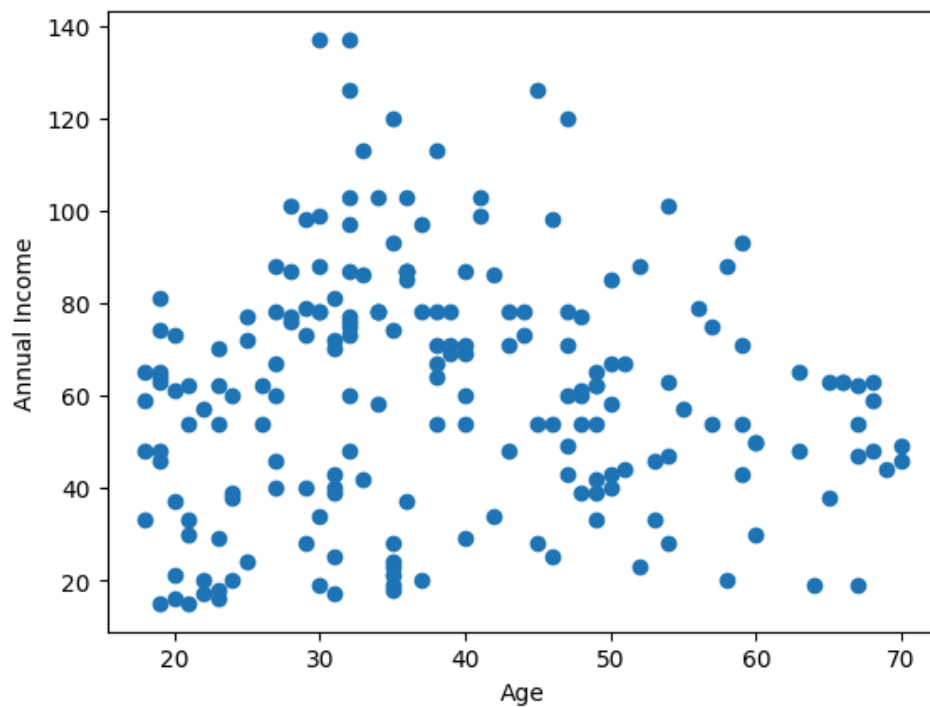
```
df.info()
```

 <class 'pandas.core.frame.DataFrame'>
 RangeIndex: 200 entries, 0 to 199
 Data columns (total 5 columns):
 # Column Non-Null Count Dtype
 --- -
 0 CustomerID 200 non-null int64
 1 Genre 200 non-null object
 2 Age 200 non-null int64
 3 Annual Income (k\$) 200 non-null int64
 4 Spending Score (1-100) 200 non-null int64
 dtypes: int64(4), object(1)
 memory usage: 7.9+ KB

```
plt.scatter(df['Age'], df['Spending Score (1-100)'])
plt.xlabel('Age')
plt.ylabel('Spending Score')
plt.show()
```



```
plt.scatter(df['Age'], df['Annual Income (k$)'])  
plt.xlabel('Age')  
plt.ylabel('Annual Income')  
plt.show()
```



```
x = df.iloc[:, [3,4]].values
```

x

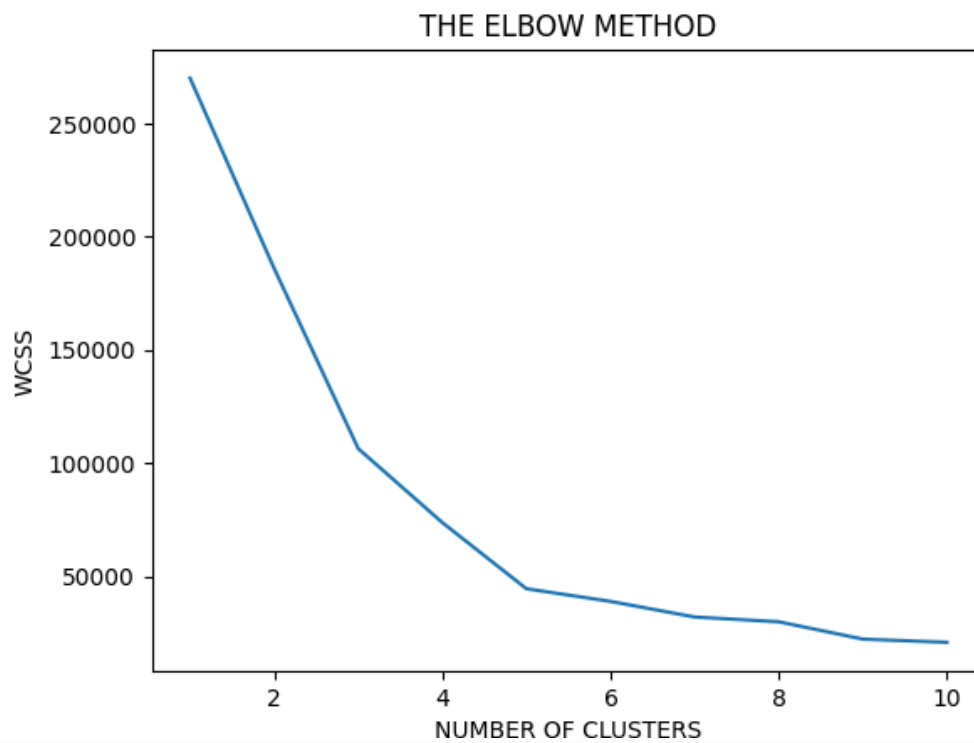


```
[ 78, 90],
[ 78, 17],
[ 78, 88],
[ 78, 20],
[ 78, 76],
[ 78, 16],
[ 78, 89],
[ 78, 1],
[ 78, 78],
[ 78, 1],
[ 78, 73],
[ 79, 35],
[ 79, 83],
[ 81, 5],
[ 81, 93],
[ 85, 26],
[ 85, 75],
[ 86, 20],
[ 86, 95],
[ 87, 27],
[ 87, 63],
[ 87, 13],
[ 87, 75],
[ 87, 10],
[ 87, 92],
[ 88, 13],
[ 88, 86],
[ 88, 15],
[ 88, 69],
[ 93, 14],
[ 93, 90],
[ 97, 32],
[ 97, 86],
[ 98, 15],
[ 98, 88],
[ 99, 39],
[ 99, 97],
[101, 24],
[101, 68],
[103, 17],
[103, 85],
[103, 23],
[103, 69],
[113, 8],
[113, 91],
[120, 16],
[120, 79],
[126, 28],
[126, 74],
[137, 18],
[137, 83]])
```

```
from sklearn.cluster import KMeans
wcss =[]
```

```
for i in range(1,11):
    kmeans=KMeans(n_clusters=i,init='k-means++',random_state=0)
    kmeans.fit(x)
    wcss.append(kmeans.inertia_)
# //kmeans.inertia is used to seggrigate the cluster
```

```
plt.plot(range(1,11),wcss)
plt.title('THE ELBOW METHOD')
plt.xlabel('NUMBER OF CLUSTERS')
plt.ylabel('WCSS')
plt.show()
```



```
kmeans_model = KMeans(n_clusters=5,init='k-means++',random_state=0)
```

```
y_kmeans = kmeans_model.fit_predict(x)
```

```
plt.scatter(x[y_kmeans==0,0],x[y_kmeans==0,1],s=80,c="red",label='Customer 1')
```

```
plt.scatter(x[y_kmeans==1,0],x[y_kmeans==1,1],s=80,c="blue",label='Customer 2')
```

```
plt.scatter(x[y_kmeans==2,0],x[y_kmeans==2,1],s=80,c="yellow",label='Customer 3')
```

```
plt.scatter(x[y_kmeans==3,0],x[y_kmeans==3,1],s=80,c="green",label='Customer 4')
```

```
plt.scatter(x[y_kmeans==4,0],x[y_kmeans==4,1],s=80,c="black",label='Customer 5')
```

```
plt.scatter(kmeans.cluster_centers_[0,0],kmeans.cluster_centers_[0,1],s=100,c='magenta',label='Centroids')
```

```
plt.title('Clusters of Customers')
```

```
plt.xlabel('Annual Income(k$)')
```

```
plt.ylabel('Spending Score(1-100)')
```

```
plt.legend()
```

```
plt.show()
```

